

How do the brain and body develop?



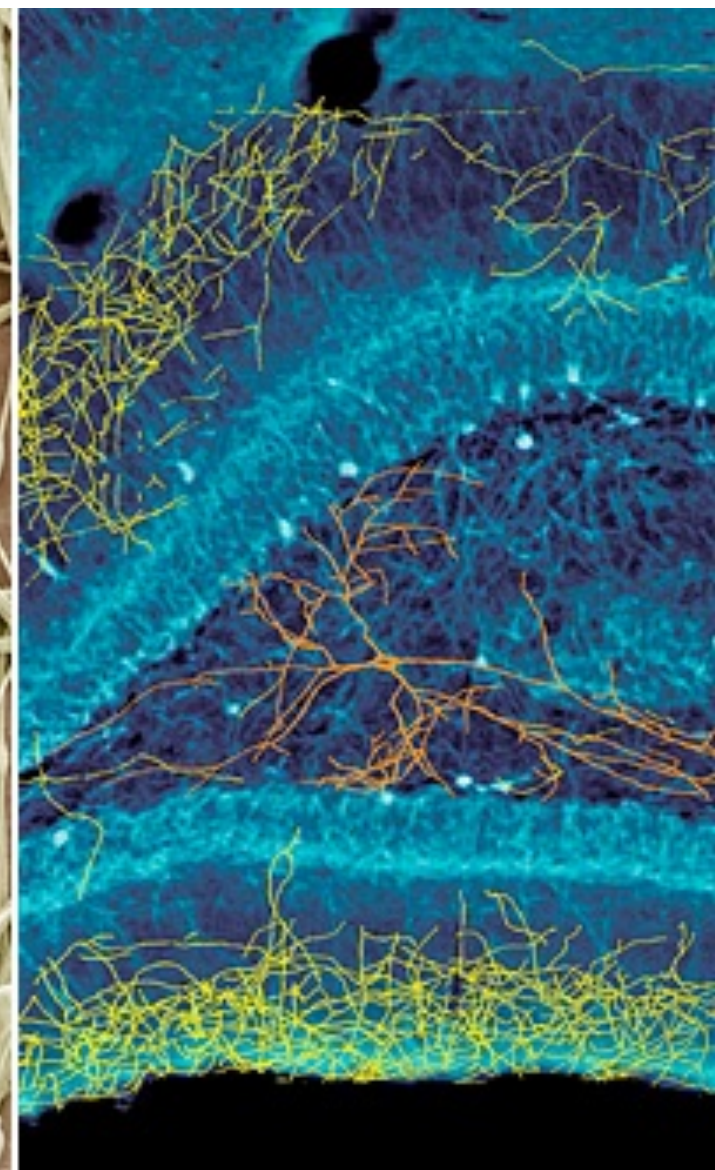
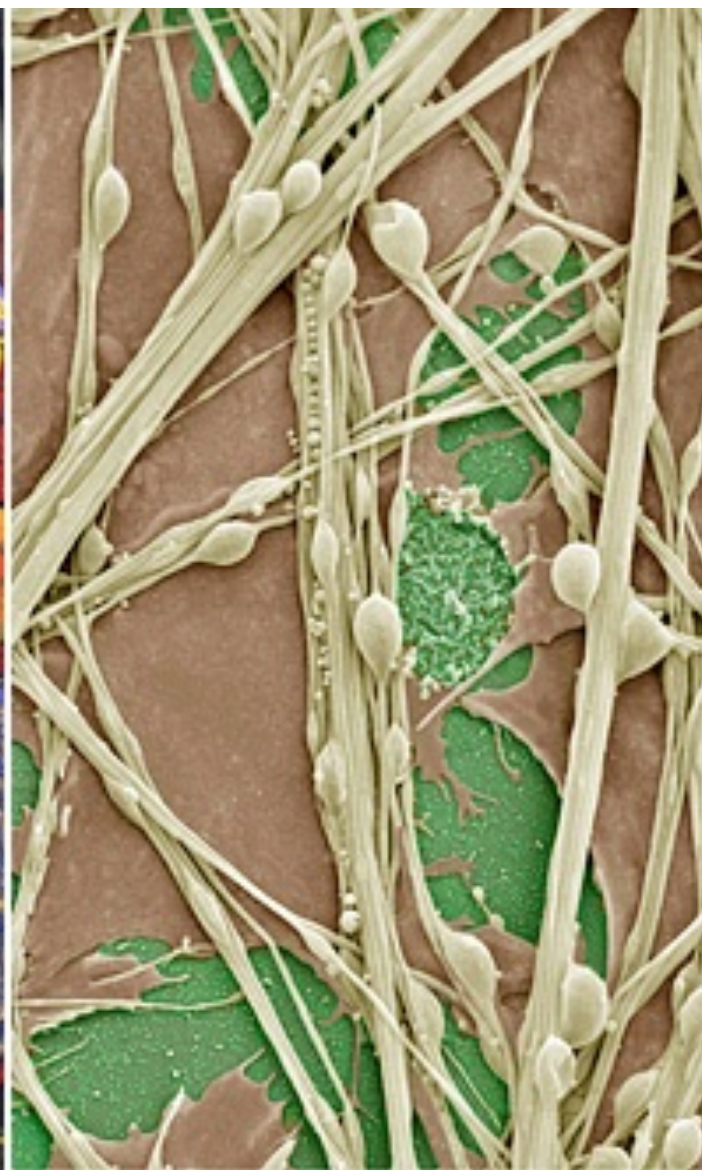
T/Th 9:35-10:25am

Professor Casey Lew-Williams

Kennedy Casey, Abby Fergus, Sarah Joo, Chantal Valdivia, Claire Whiting

Today

- Brain development
 - Structures of the brain
 - Developmental processes
- Physical development
 - Nutrition
 - Sleep



*Portraits of the Mind:
Visualizing the Brain from Antiquity to the 21st Century*

brain DEVELOPMENT

by DWAYNE GODWIN AND JORGE CHAM

THERE ARE OVER 100 BILLION NEURONS IN YOUR BRAIN...

(THE SAME NUMBER AS THERE ARE STARS IN THE MILKY WAY GALAXY)

...AND EACH NEURON CONNECTS TO ABOUT 10,000 OTHER NEURONS.

THAT'S A QUADRILLION CONNECTIONS, OR
1,000,000,000,000,000!
(THE SAME NUMBER OF CELLS YOU HAVE IN YOUR ENTIRE BODY)

HOW DOES SUCH A COMPLEX NETWORK DEVELOP??

WELL, IT ALL STARTS WHEN YOU'RE ABOUT 4 WEEKS OLD...

YOU.

A NARROW GROOVE BEGINS TO FORM ON YOUR NASCENT GROUP OF CELLS, GIVING THE FIRST HINT OF A NERVOUS SYSTEM:

THE GROOVE THEN FOLDS INTO ITSELF, FORMING WHAT IS CALLED THE NEURAL TUBE.

THE TUBE THEN SQUEEZES IN VARIOUS SECTIONS TO FORM THE DIFFERENT PARTS OF THE BRAIN AND THE SPINAL CORD:



NEW NEURONS ARE MADE IN AN AREA DEEP INSIDE THE GROWING BRAIN CALLED THE VENTRICULAR ZONE...

...AND THEN TRAVEL OUTWARDS BY CLIMBING ALONG CELLULAR "ROPE" CALLED "BERGMANN GLIA".

THEY EVENTUALLY SETTLE INTO SIX DISTINCT LAYERS OF NEURONS THAT MAKE UP YOUR BRAIN CORTEX.



ONCE IN PLACE, NEURONS REACH OUT TO EACH OTHER BY EXTENDING TENTACLE-LIKE "GROWTH CONES"...

BUT IT'S NOT ALL GENETIC.

SYNCHRONIZED ACTIVATION OF TWO NEURONS REINFORCES THEIR CONNECTION...

THIS IS WHY ALCOHOL IS SO HARMFUL TO DEVELOPING BRAINS.

IT NOT ONLY DISRUPTS THE GENES INVOLVED IN DEVELOPMENT, BUT ALSO AFFECTS THE CHEMICALS THAT TRANSMIT SIGNALS BETWEEN NEURONS.

...WHICH ARE GUIDED BY CHEMICAL CUES THAT LURE AND REPEL THEM TOWARDS RECEPTORS IN OTHER NEURONS.

NEURON CONNECTIONS ARE ALSO DETERMINED BY ACTIVITY PATTERNS, WHICH IS WHAT MAKES EACH BRAIN SO UNIQUE.

...WHEREAS LACK OF USE TENDS TO DISCOURAGE CONNECTIONS.



EVEN AFTER BIRTH, OUR BRAINS ARE STILL IN THE PROCESS OF WIRING THEMSELVES AND CAN BE INFLUENCED BY THE WORLD AROUND US.

FOR EXAMPLE, CATS RAISED IN AN ENVIRONMENT OF VERTICAL STRIPES...

...LATER HAVE TROUBLE SEEING HORIZONTAL STRIPES!

ALSO, DIFFERENT PARTS OF THE BRAIN DEVELOP AT DIFFERENT RATES.

FOR EXAMPLE, THE FRONTAL LOBES (WHERE JUDGEMENT AND DECISION-MAKING HAPPENS) ARE STILL IN DEVELOPMENT DURING OUR TEENAGE YEARS...

...WHICH MIGHT EXPLAIN A LOT.

YOU MIGHT NOT FEEL ANY SMARTER, BUT YOUR BRAIN IS STILL GROWING AND REARRANGING ITSELF, EVEN AS AN ADULT (THOUGH NOT AS FAST AS IN THE WOMB)

WHAT WERE YOU WEARING?!

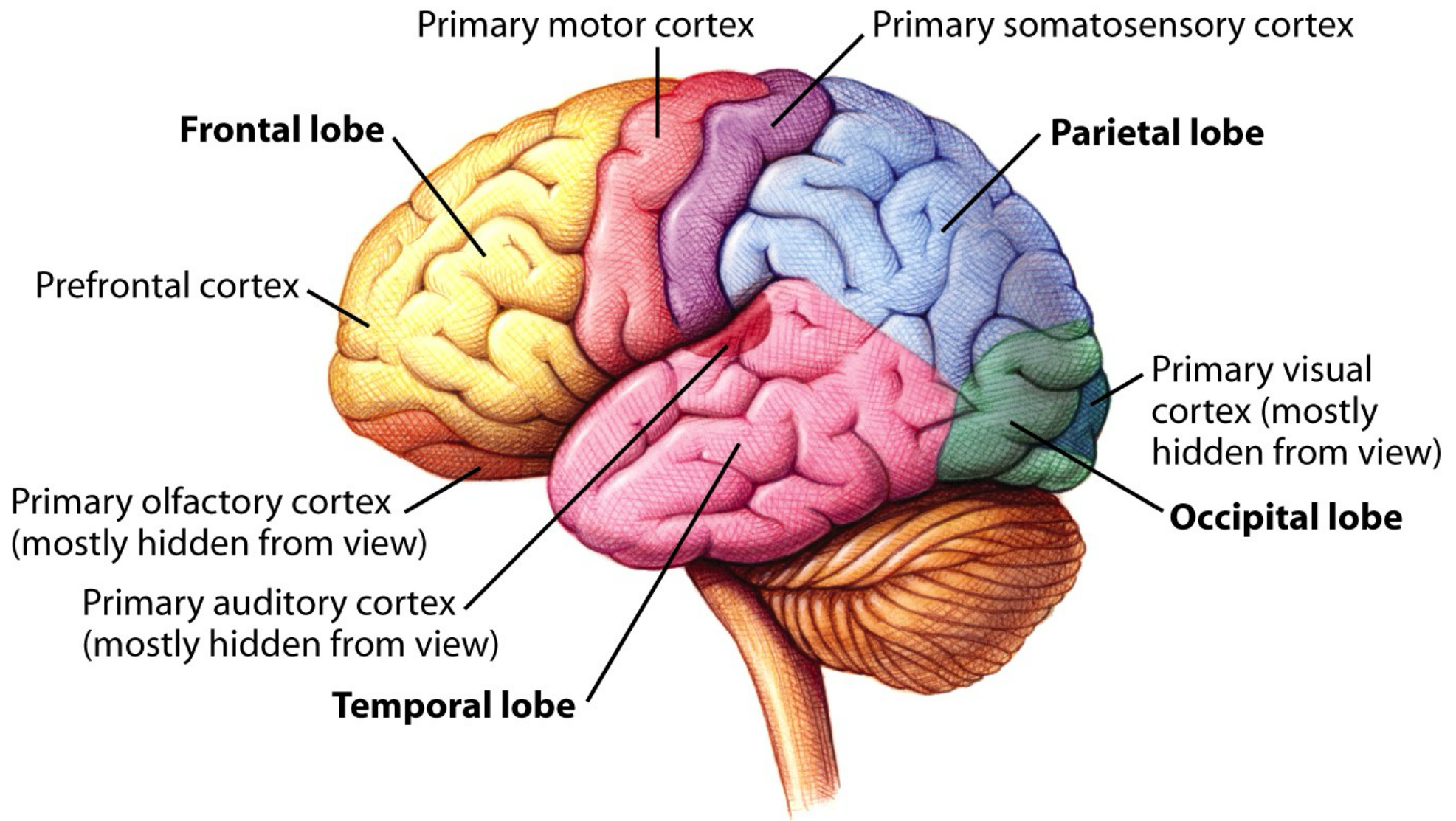


SO REMEMBER TO MAINTAIN AN ELEMENT OF SURPRISE AND WONDER ABOUT THE WORLD AROUND YOU. YOUR BRAIN NEEDS IT!



THE TEENAGE BRAIN

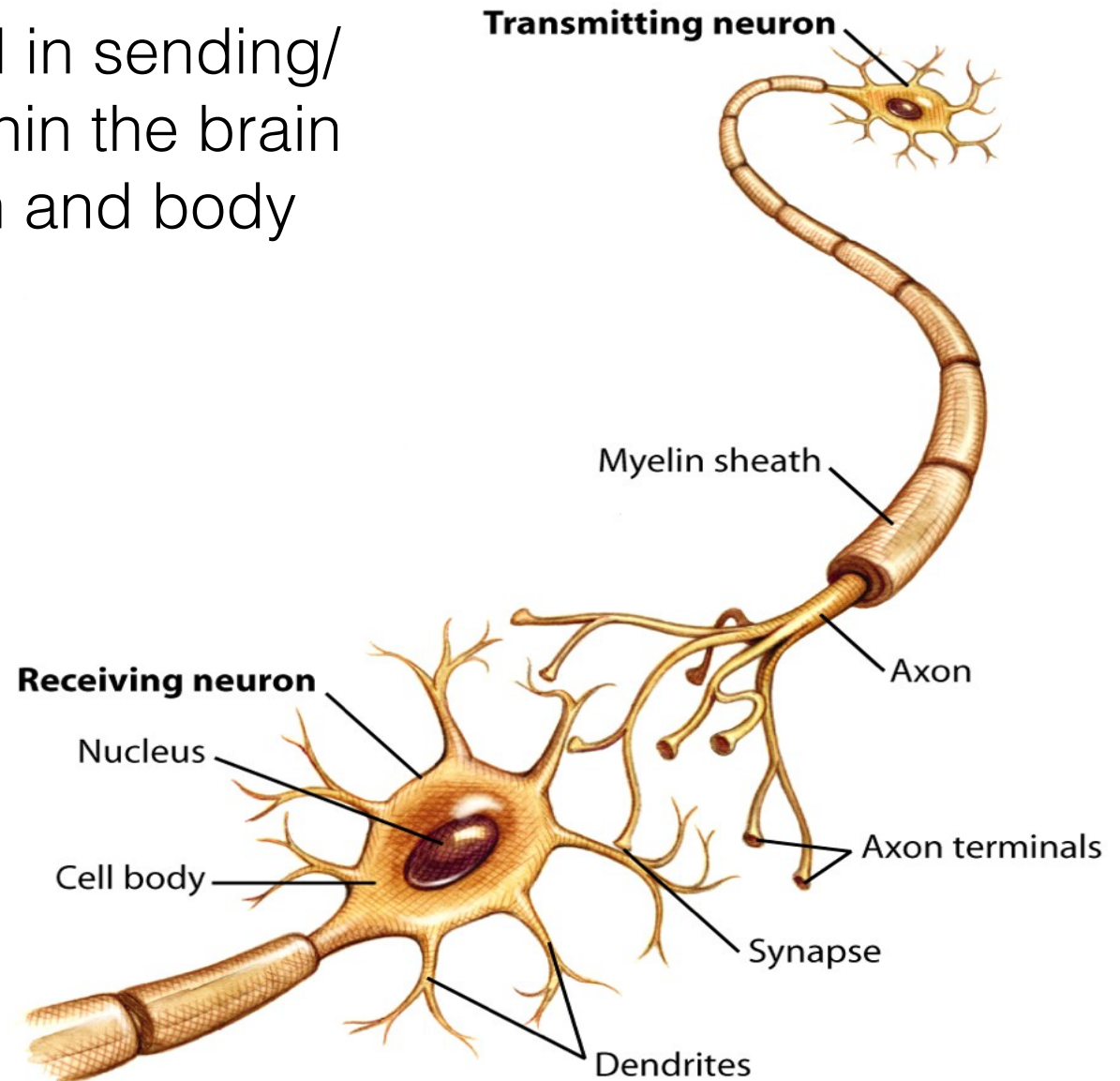
Cerebral cortex: “the most human part of the human brain”



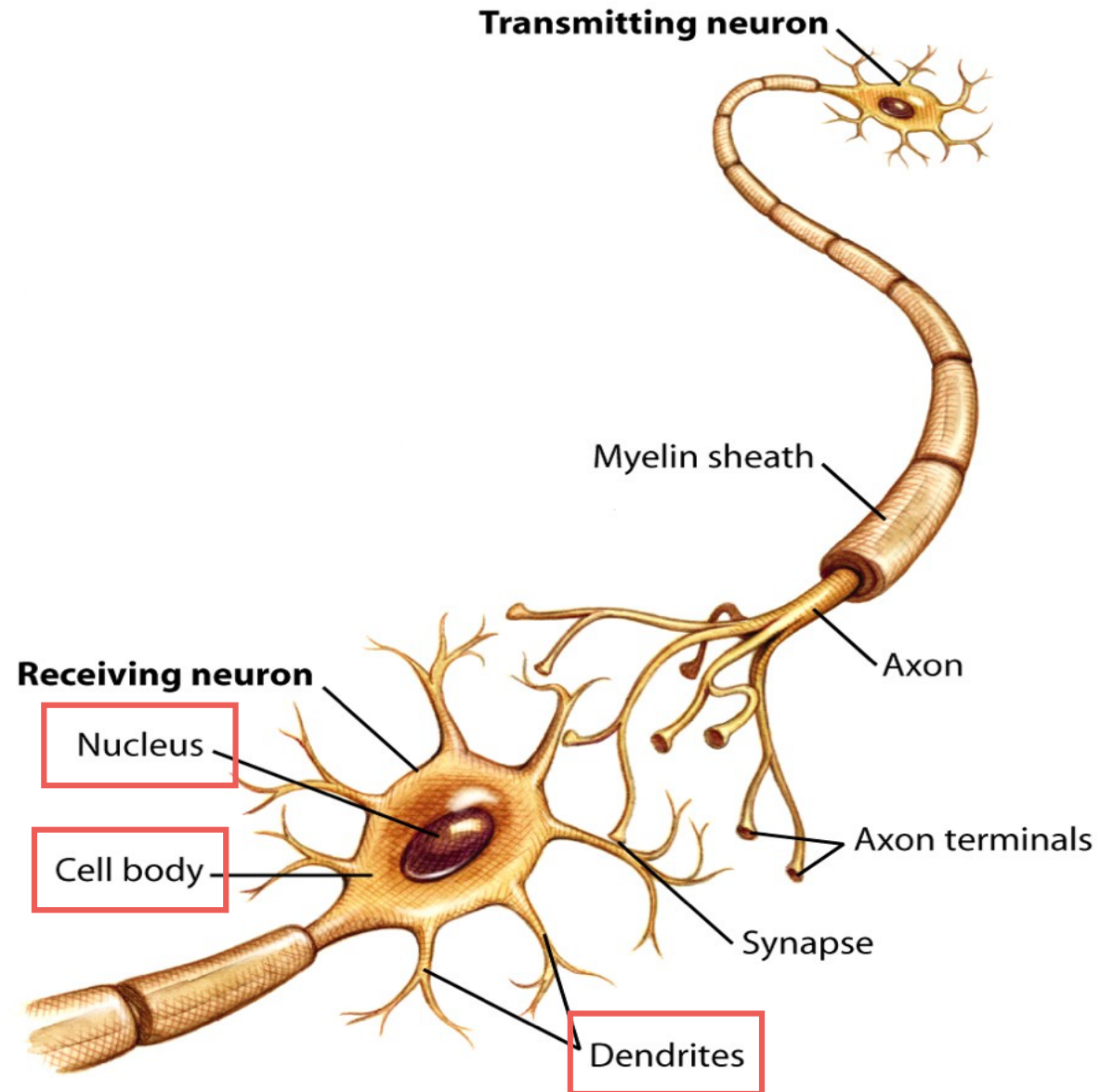
Brain structures

Neuron: cell specialized in sending/receiving messages within the brain and between the brain and body

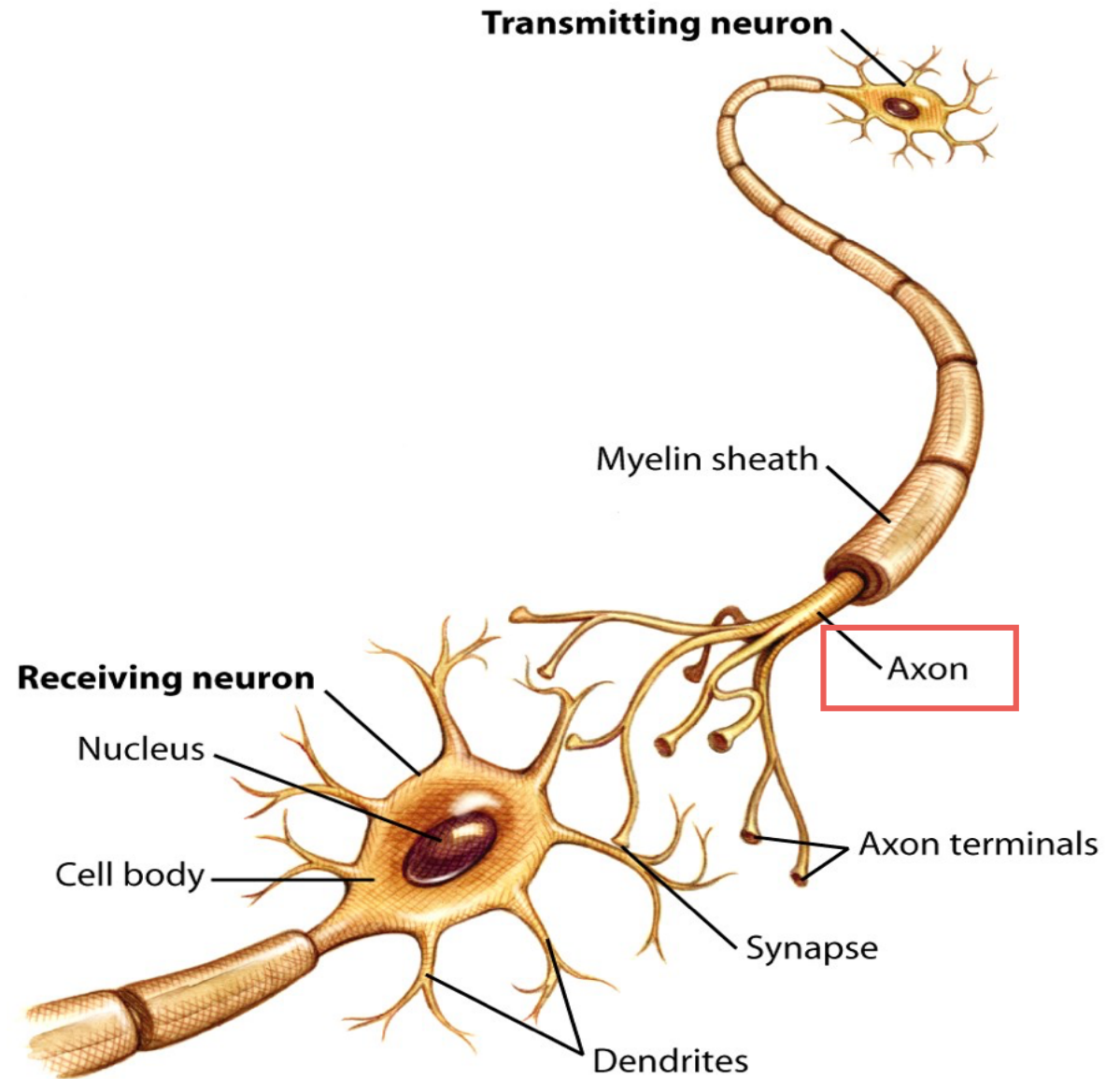
~80 billion!



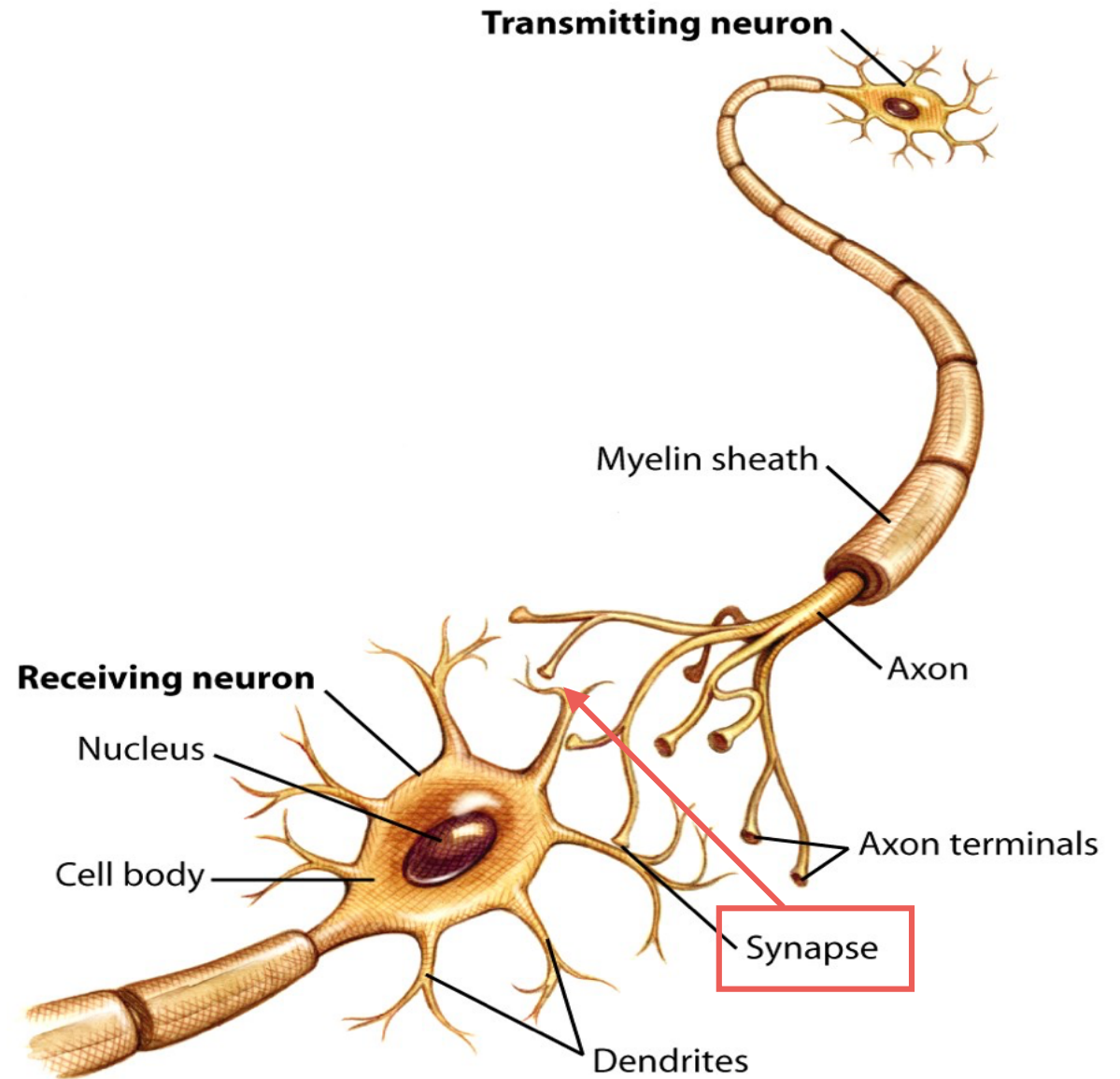
Brain structures



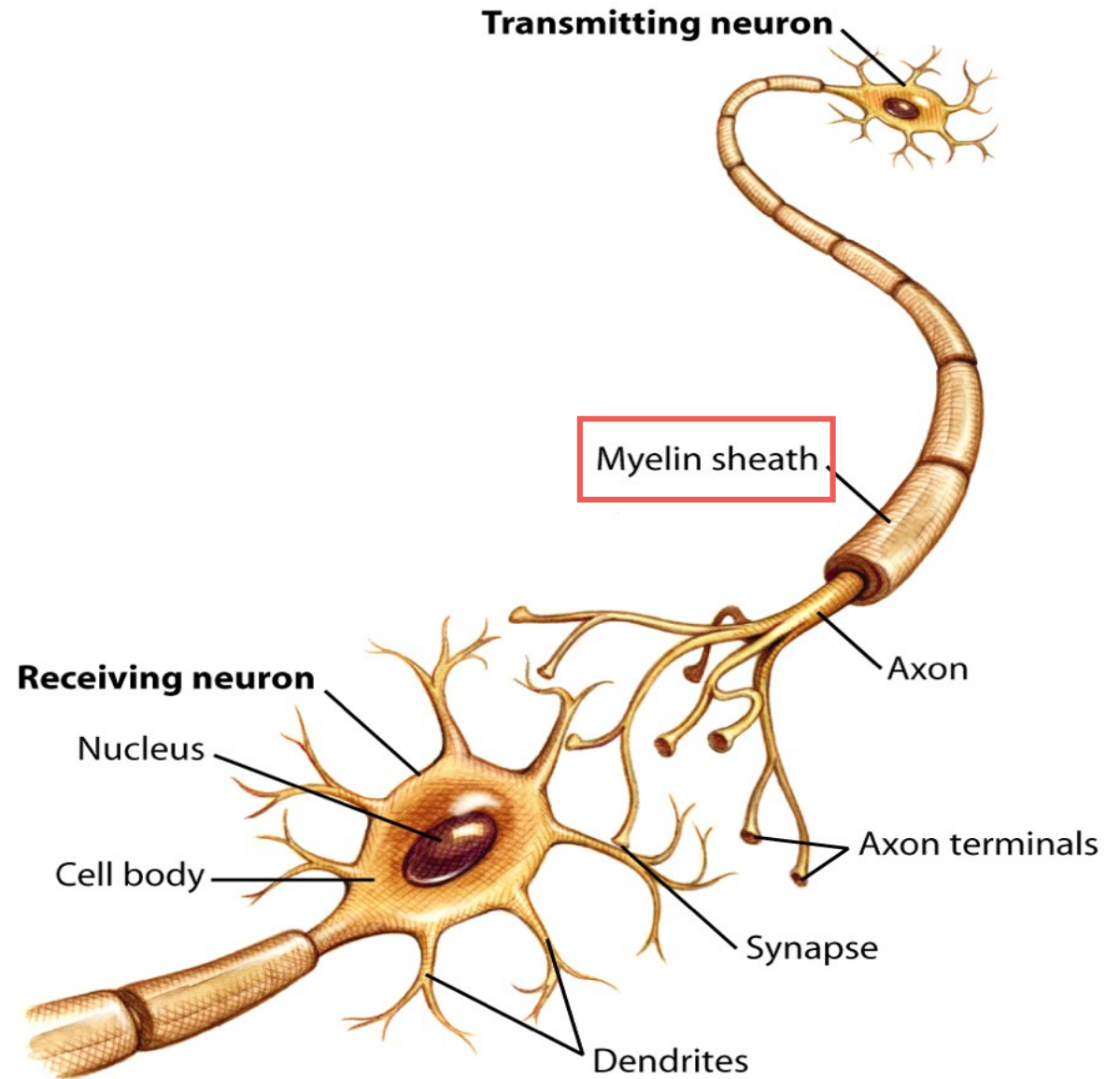
Brain structures



Brain structures



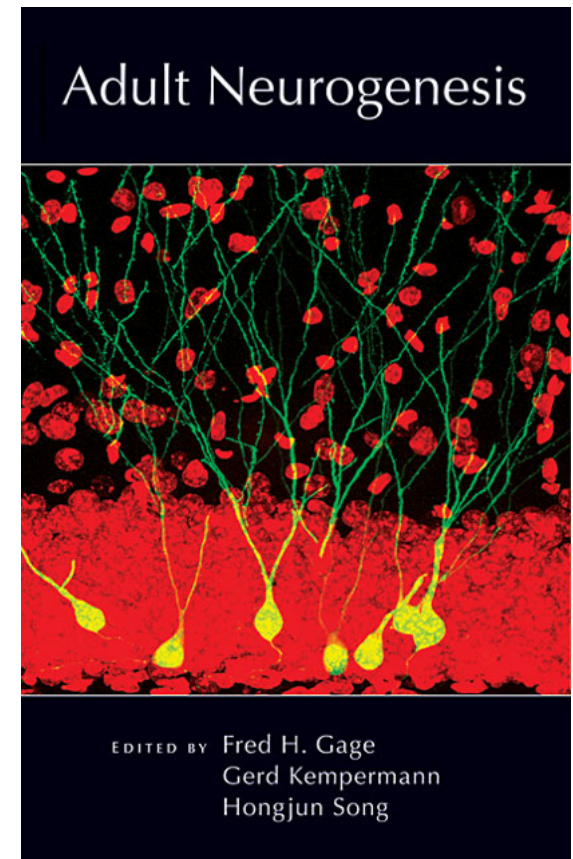
Brain structures



Developmental processes

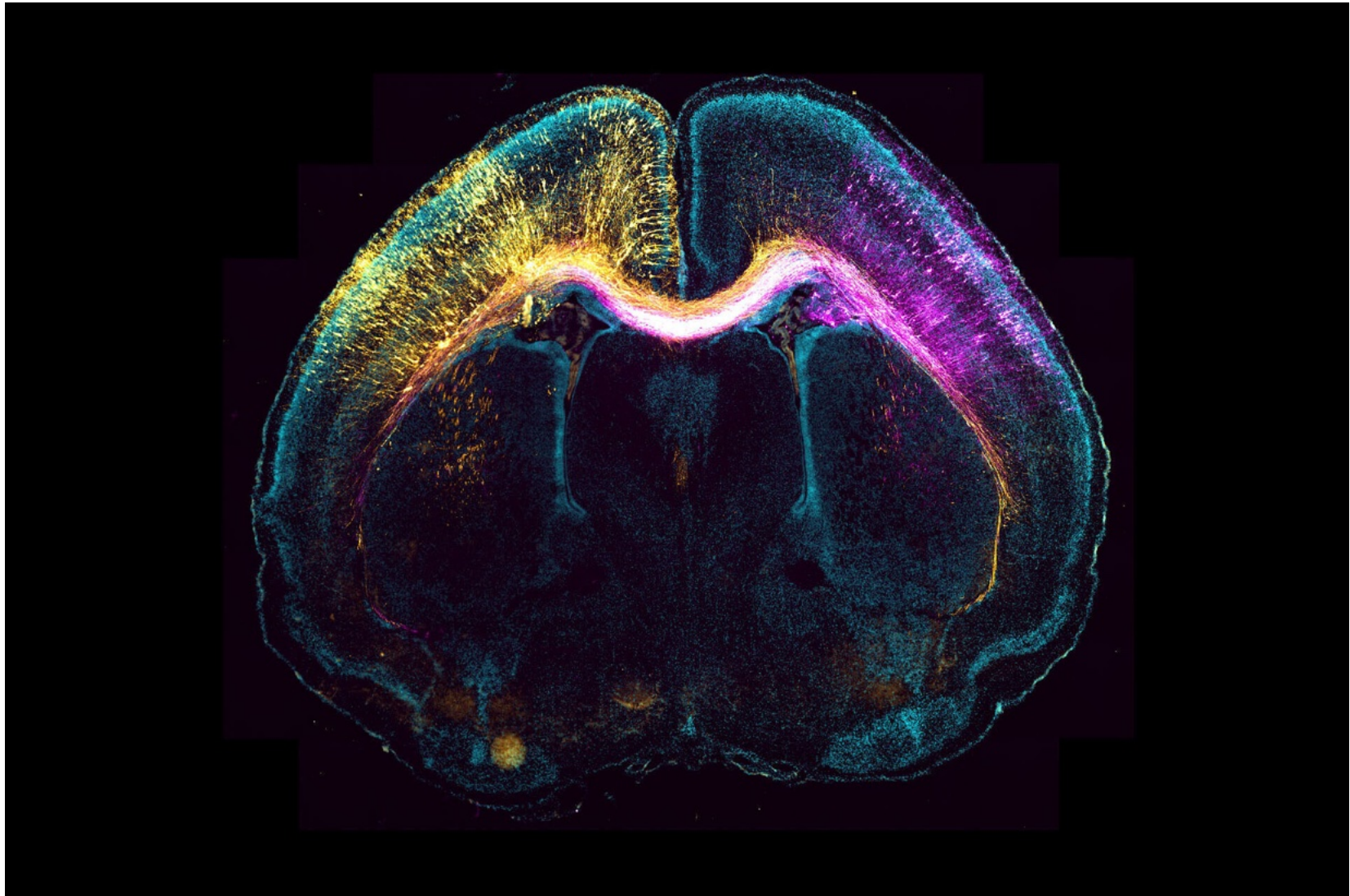
1. Neurogenesis

- Production of cells through cell division
- Most cells are created prenatally
- At birth, you have most of your neurons, but few connections.



Developmental processes

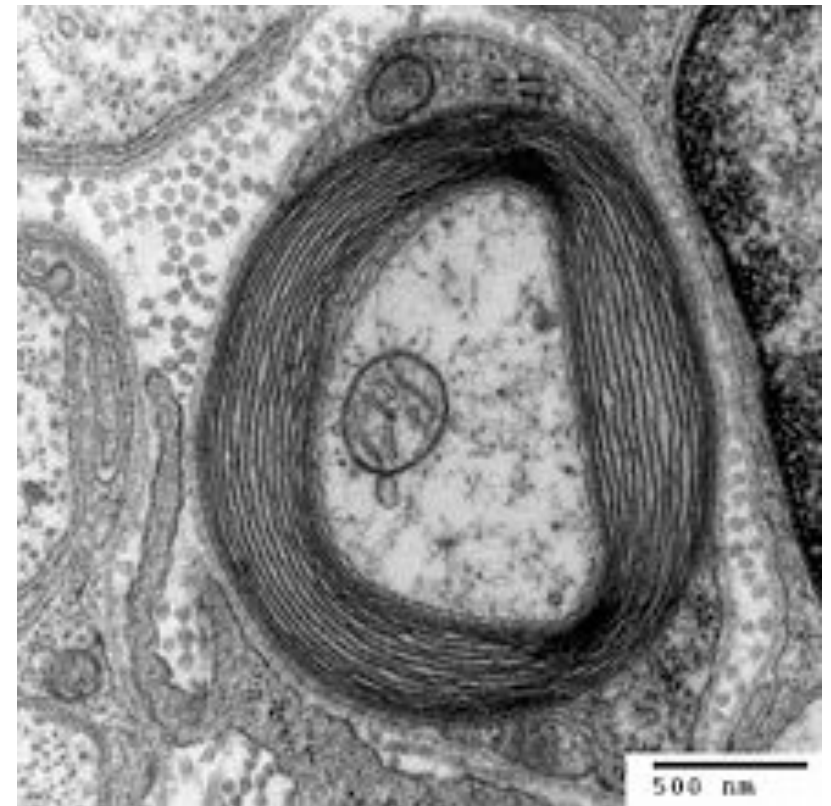
2. Migration: neurons move to appropriate locations.



Developmental processes

3. Myelination

- Formation of fatty sheath.
- Starts 4 months after conception



Developmental processes

4. Synaptogenesis

- Formation of connections so that the right cells can communicate.

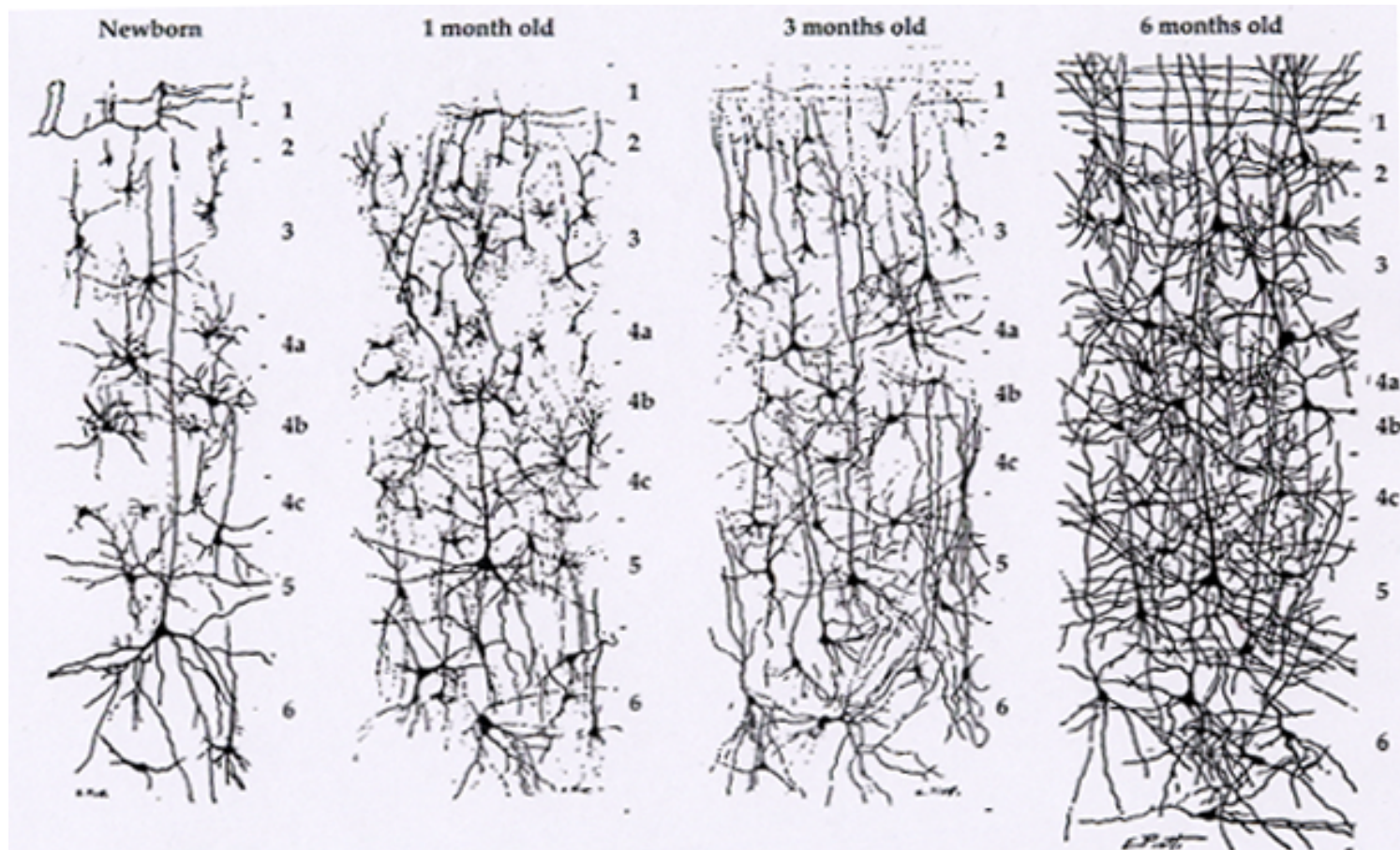
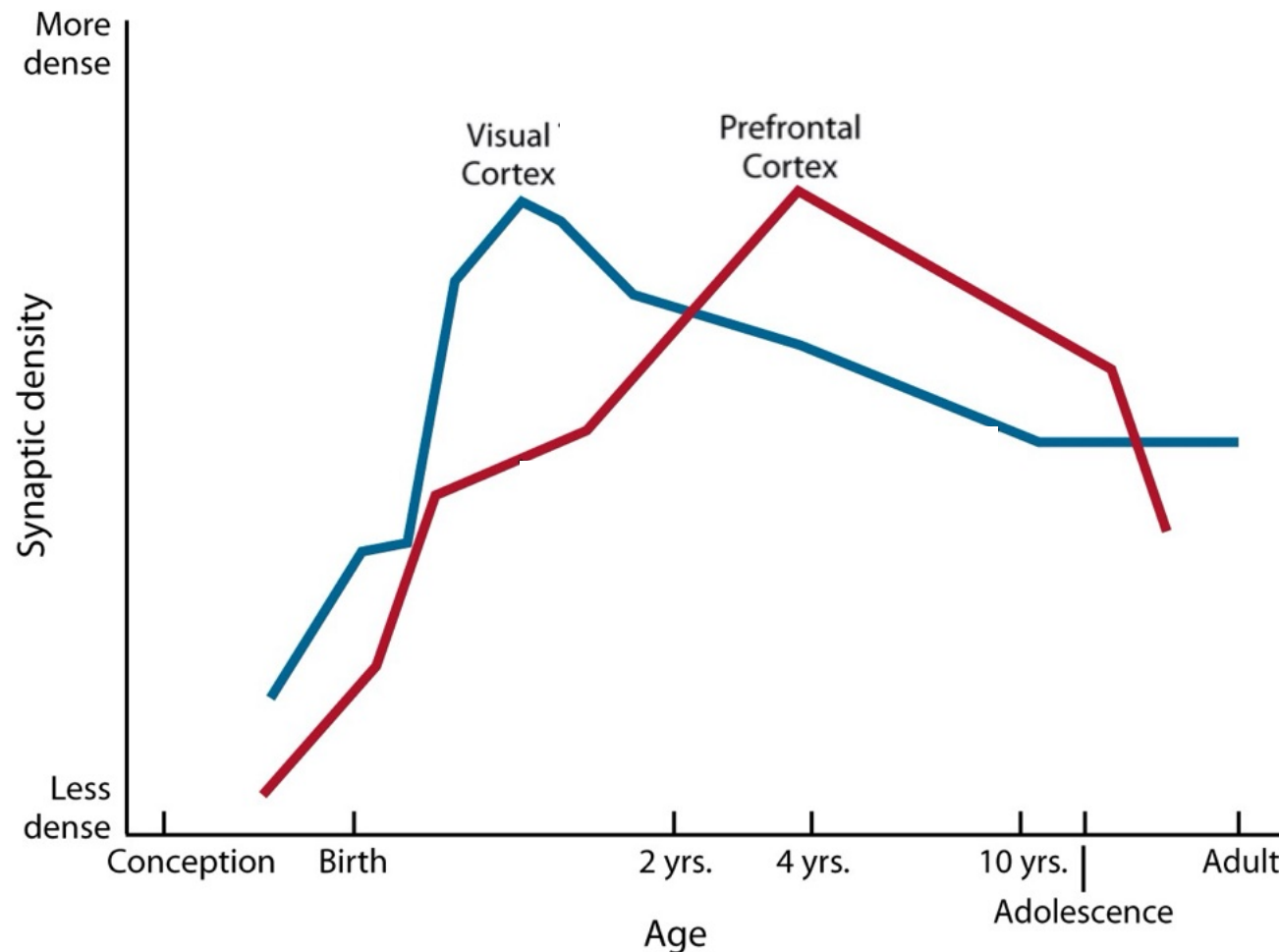


Figure 2.4 A drawing of the cellular structure of the human visual cortex based on Golgi stain preparations from Conel (1939-67).

Developmental processes

5. Synaptic pruning

- Rarely used synapses are eliminated

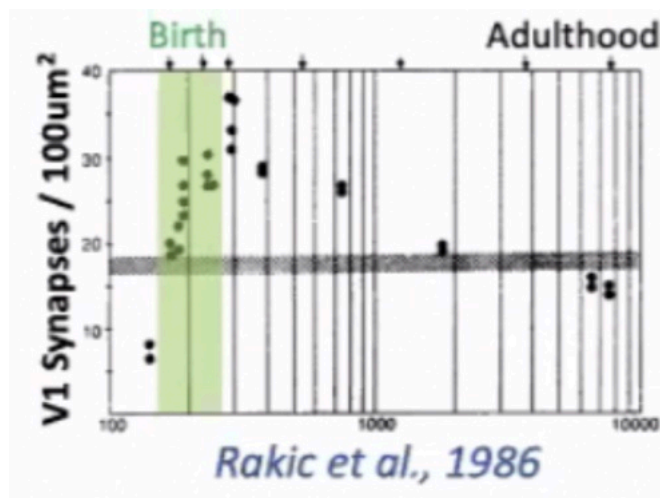


Two competing developmental theories

Pruning

Early brain development is driven by pruning of unused/unnecessary connections.

We increase responsivity in pathways that matter by reducing responsivity to irrelevant stimuli.

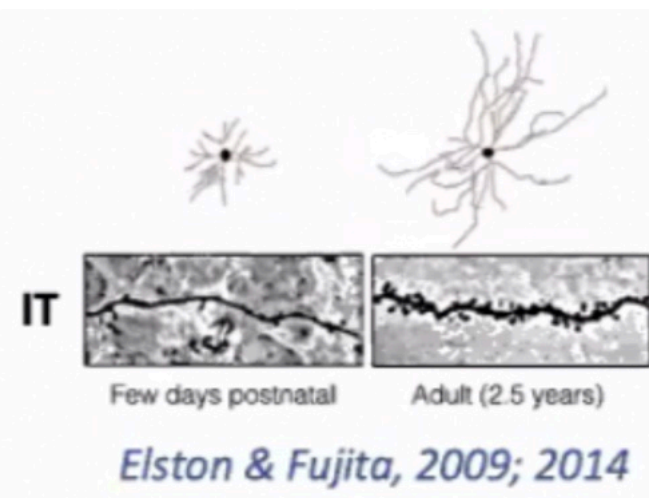


vs.

Growth

Early brain development is driven by growth of dendritic arbors, synapses, and connections.

This growth increases responsivity to relevant stimuli.



We have no idea which one is right!

Plasticity

The extent to which brain organization is flexible and shaped by experience



Experience-dependent plasticity

- Process through which individual experiences lead to creation/reorganization of neural connections.
- Example: rats raised in complex environments
- Example: violinists
- Example: Prof. Jesse Gomez's Pokémon paper in your precept!



Experience-expectant plasticity

- Process through which typical brain wiring occurs, resulting from the typical human environment, available throughout human evolution.
- Brain “expects” certain experiences (e.g., visual patterns, movement, human voices?) and needs input to fine-tune the system.
- Sensitive periods: brain is especially sensitive to particular input during a particular window. If input isn’t available during that period, development is less likely to be typical. (it’s not all-or-nothing).

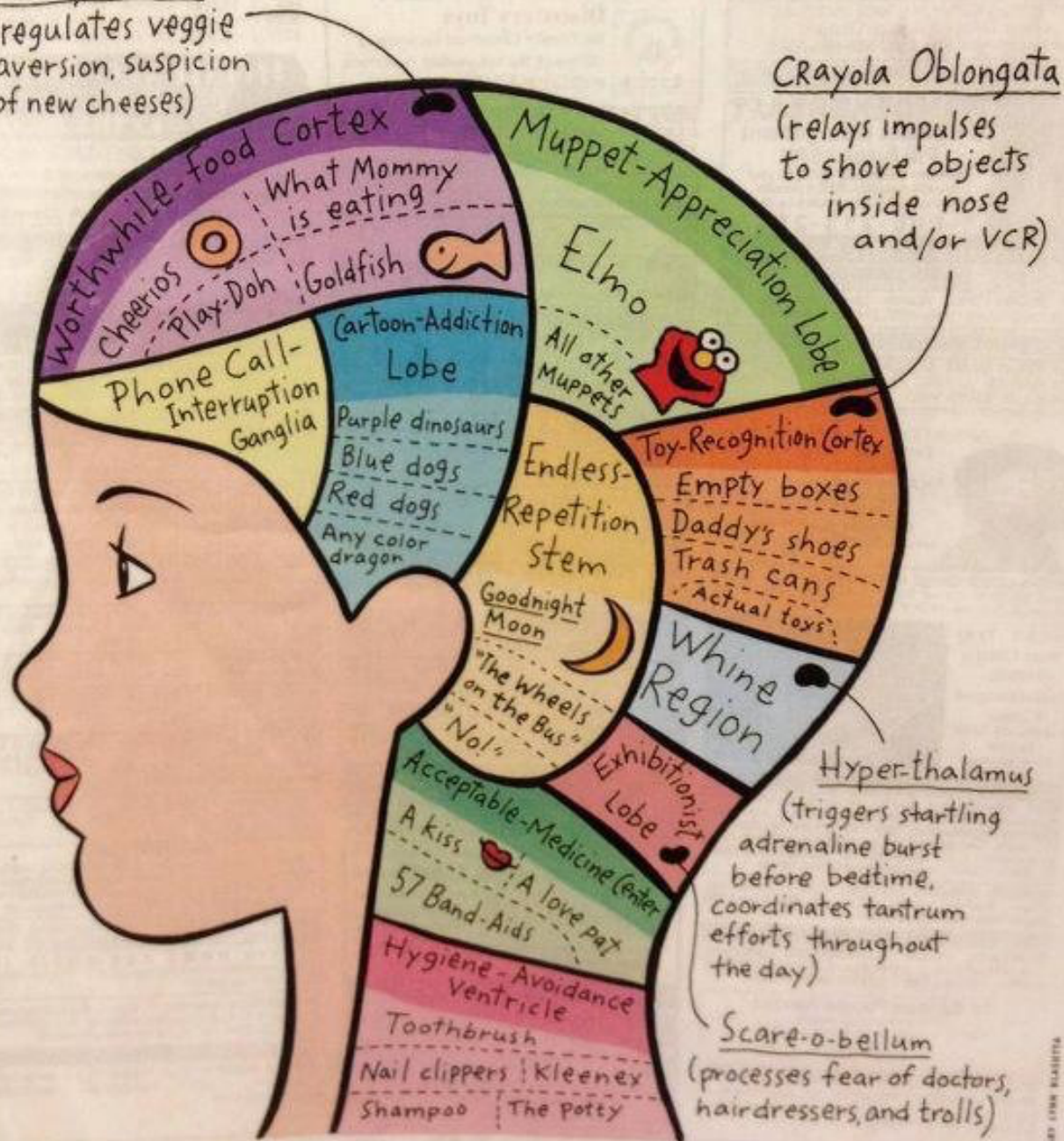
Example: strabismus



Inside a toddler's brain

Ptooley Gland

(regulates veggie aversion, suspicion of new cheeses)



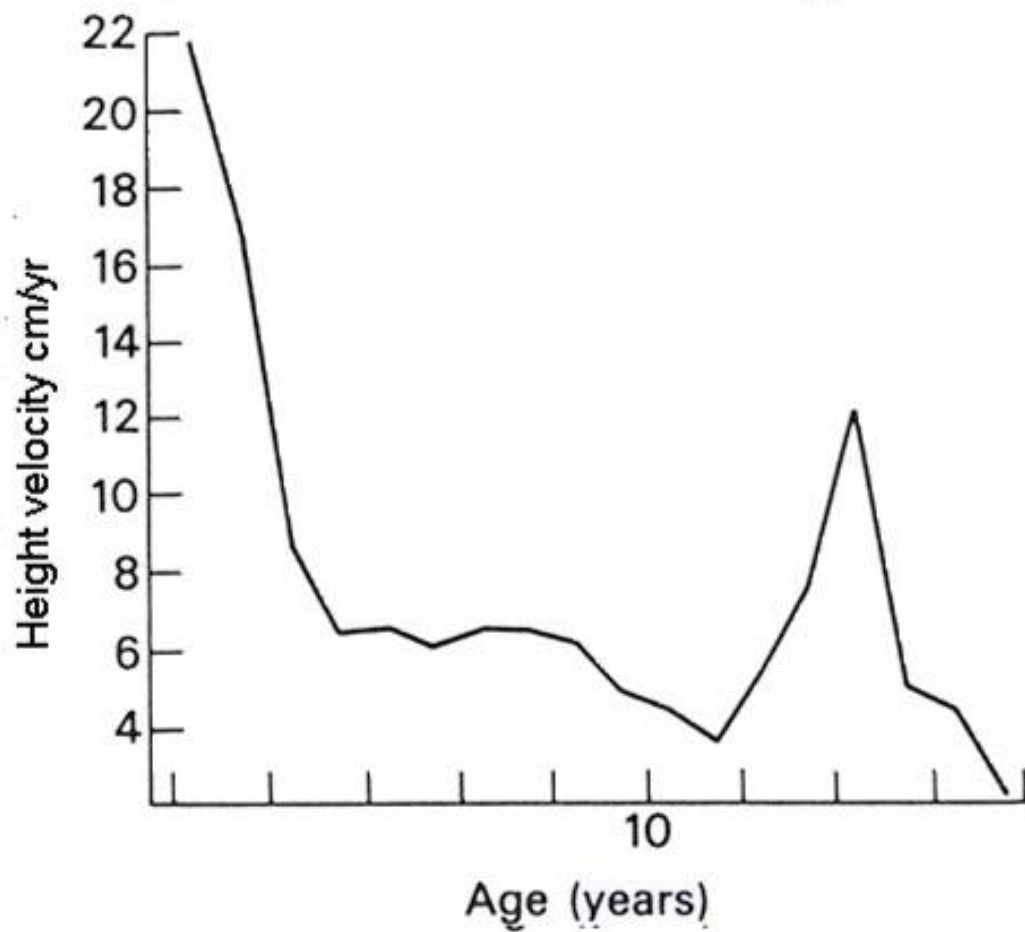
-MELISSA BALMAIN

HEAD (TOP) BLANKET

Plasticity +
Plasticity -



Growth rates





Mechanisms of physical development



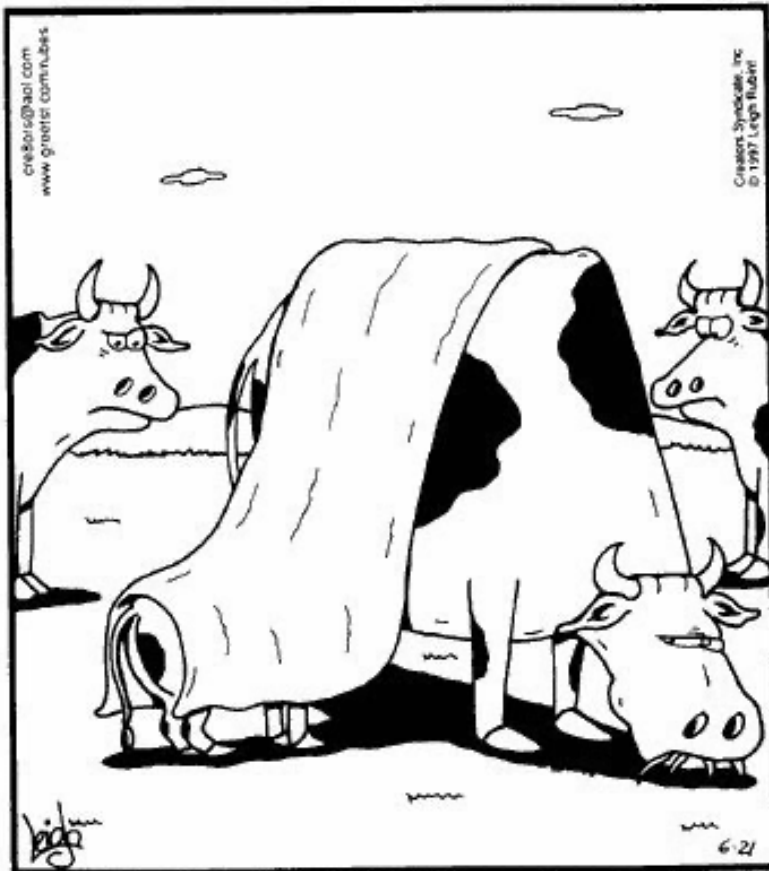
Nutrition

- Babies need lots of food
- Formula vs. breastfeeding
- Formula in developing countries —> sick babies. Why?

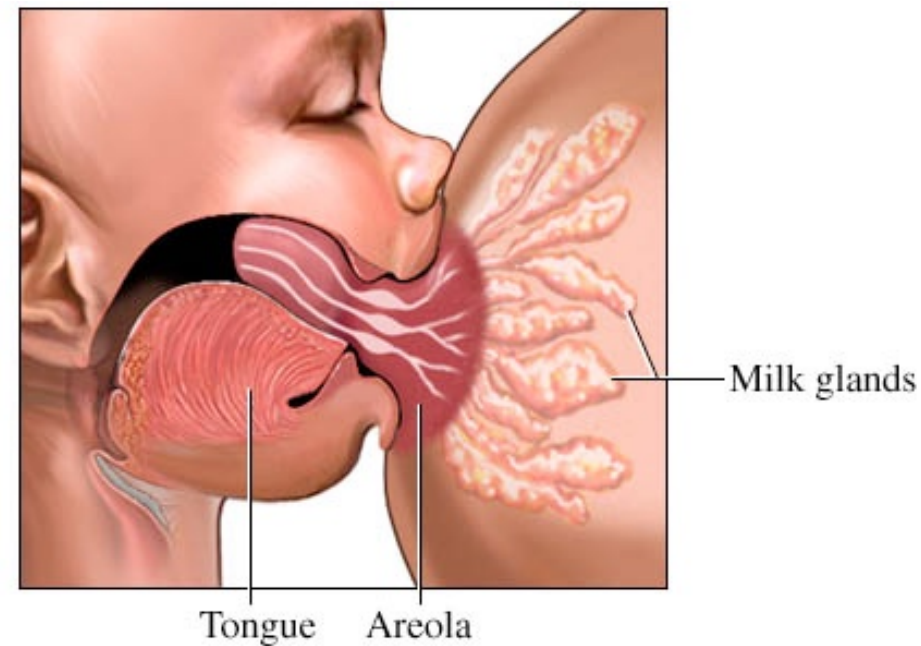


Earliest nutrition: Breastfeeding

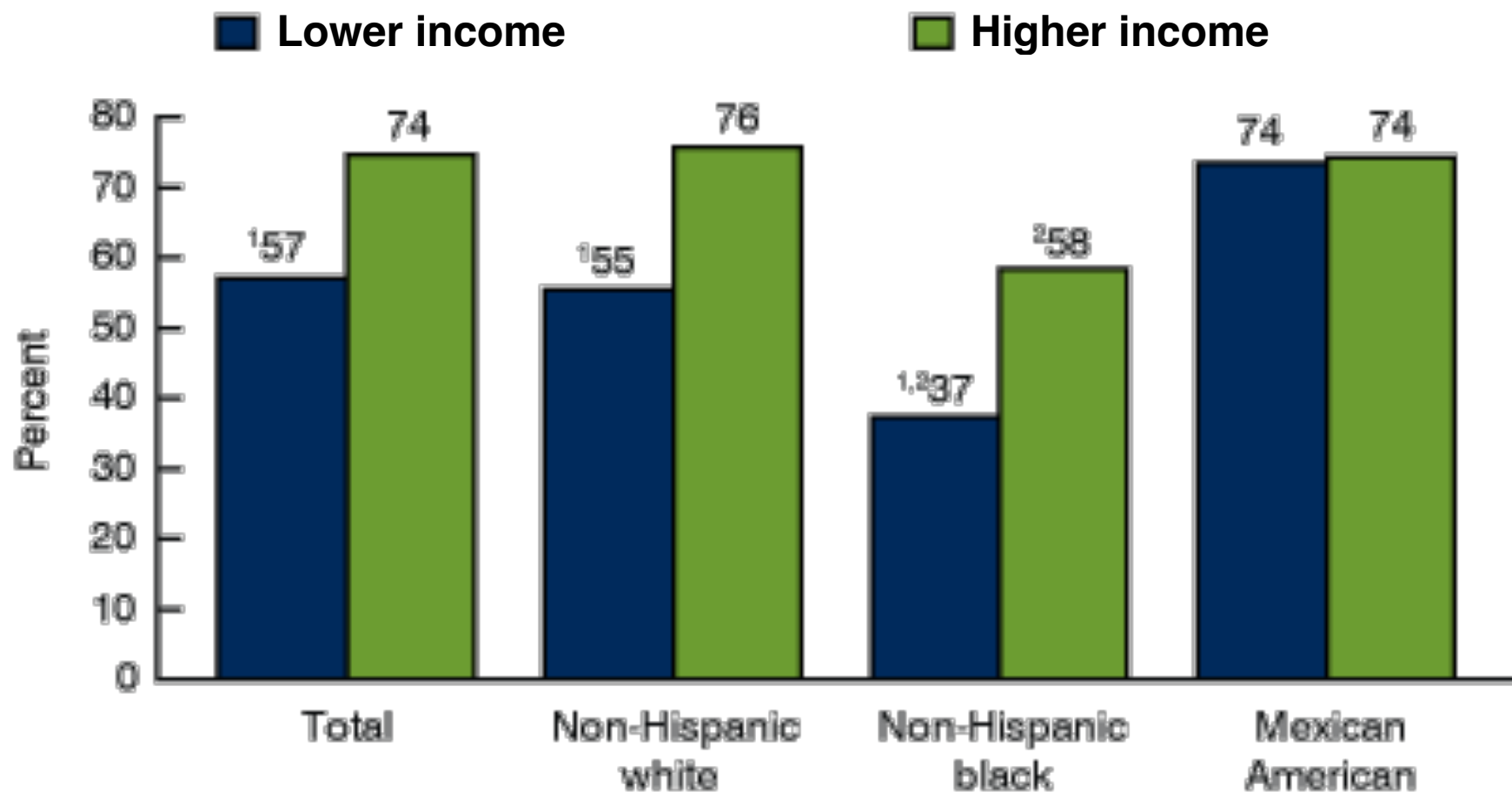
- Health
- Immune system
- Promotes myelination
- Lower risk of obesity throughout life
- Optimal for brain development



Even when practiced discreetly, raised eyebrows and disapproving glances still meet with those who dare to udder-feed in public.



Earliest nutrition: % breastfeeding



¹Significant differences between the two income groups.

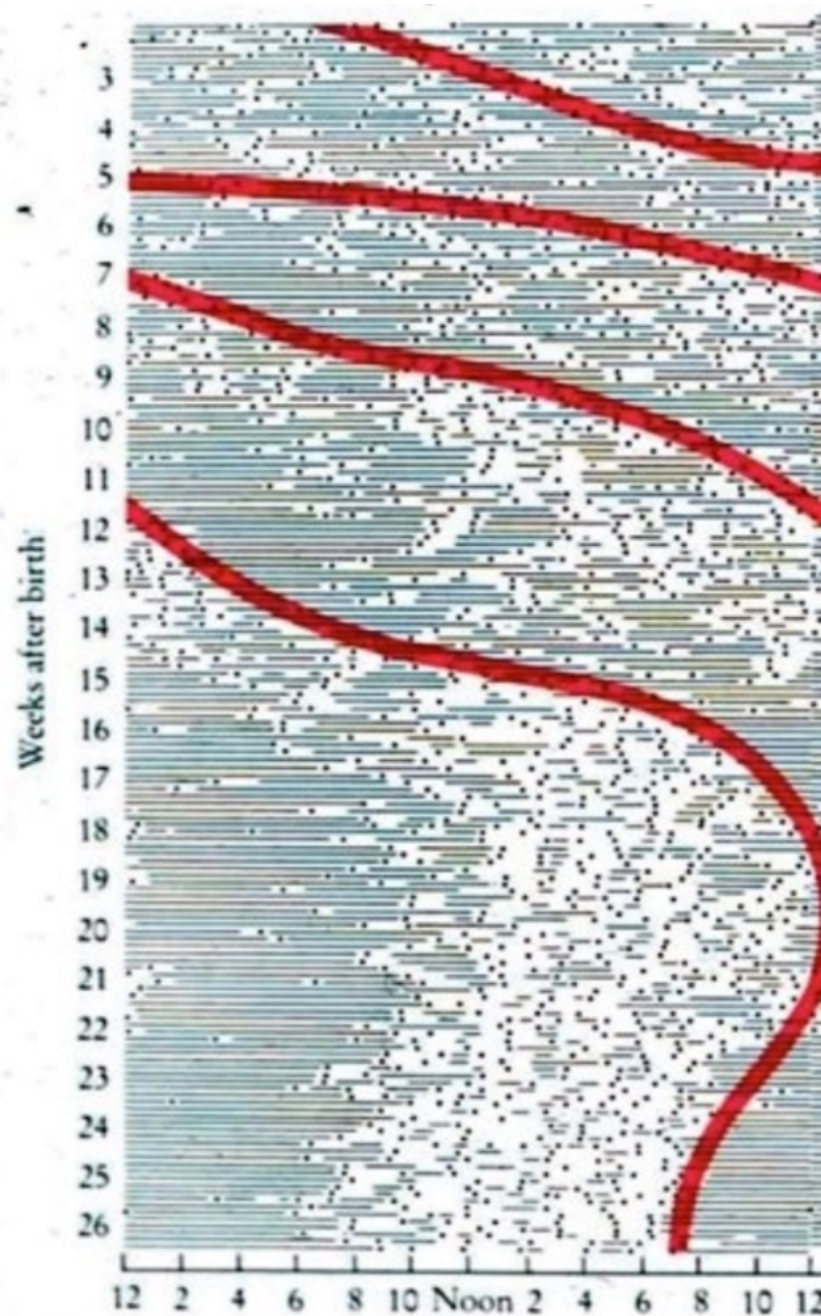
²Significantly different from non-Hispanic white and Mexican-American infants within income groups.

Developmental mechanism: Sleep

- Why does quality of sleep matter?
 - Physical: ~80% of growth hormones secreted during sleep
 - Cognitive: children with sufficient sleep have better attention, academic performance, fewer behavioral problems.

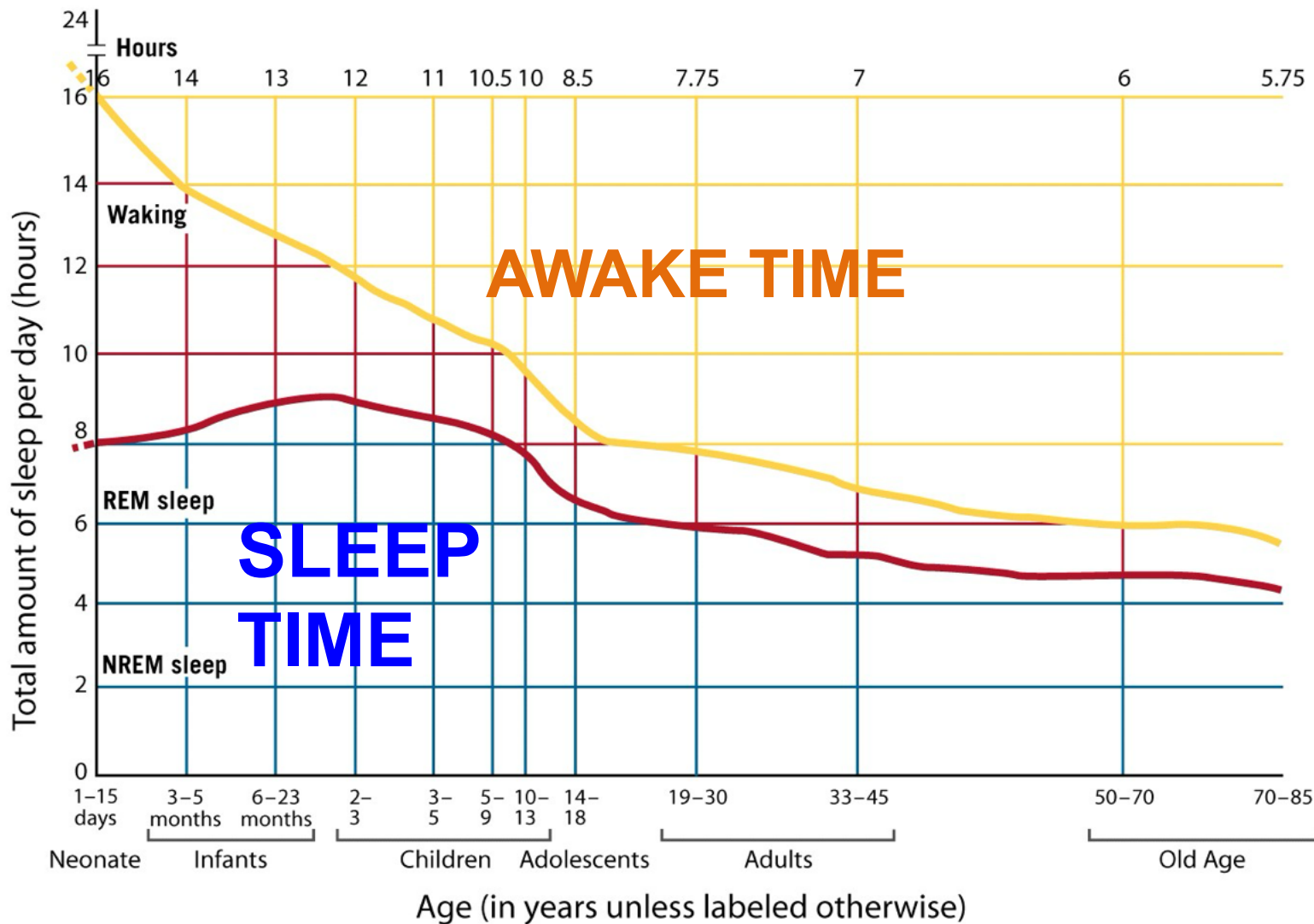


Newborn becomes synchronized with the light/dark cycle



■ nursing — sleeping

Sleep over the course of development



Recommended sleep
Newborns: 15-17 hours
1-12 mos: 14-15 hours
1-3 years: 12-14 hours
3-12: 10-11/12 hours
12-18: 8.5-9.6 hours
Adults: 7-9 hours

Sleep and adolescence

- Teens need 9 hours of sleep; get ~7 hours
- Changes in sleep patterns: late to bed, late to rise
- Maybe it wasn't your fault? Biological changes in circadian rhythms.
- Melatonin



Should high school start later?

- Minneapolis: high school start was moved from 7:15 to 8:40
- Researchers found:
 - Students report: less depression, less irritability
 - Teacher report: teens more alert, on time, less sleepy in class
 - Parent report: teens easier to live with.
 - Lower dropout rates
 - Fewer car accidents



Thank you, Campus Dining

